



Micro Commercial Components



Micro Commercial Components
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MMS8050-L

MMS8050-H

Features

- SOT-23 Plastic-Encapsulate Transistors
- Capable of 0.3Watts($T_{amb}=25^{\circ}\text{C}$) of Power Dissipation.
- Collector-current 0.5A
- Collector-base Voltage 40V
- Operating and storage junction temperature range: -55°C to $+150^{\circ}\text{C}$
- Marking Code: J3Y
- Lead Free Finish/RoHS Compliant ("P" Suffix designates RoHS Compliant. See ordering information)
- Epoxy meets UL 94 V-0 flammability rating
- Moisture Sensitivity Level 1

Electrical Characteristics @ 25°C Unless Otherwise Specified

Symbol	Parameter	Min	Max	Units
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OFF CHARACTERISTICS

$V_{(BR)CBO}$	Collector-Base Breakdown Voltage ($I_C=100\mu\text{Adc}$, $I_E=0$)	40	---	Vdc
$V_{(BR)CEO}$	Collector-Emitter Breakdown Voltage ($I_C=0.1\text{mAdc}$, $I_B=0$)	25	---	Vdc
$V_{(BR)EBO}$	Emitter-Base Breakdown Voltage ($I_E=100\mu\text{Adc}$, $I_C=0$)	5.0	---	Vdc
I_{CBO}	Collector Cutoff Current ($V_{CB}=40\text{Vdc}$, $I_E=0$)	---	0.1	μAdc
I_{CEO}	Collector Cutoff Current ($V_{CE}=20\text{Vdc}$, $I_B=0$)	---	0.1	μAdc
I_{EBO}	Emitter Cutoff Current ($V_{EB}=5.0\text{Vdc}$, $I_C=0$)	---	0.1	μAdc

ON CHARACTERISTICS

$h_{FE(1)}$	DC Current Gain ($I_C=50\text{mAdc}$, $V_{CE}=1.0\text{Vdc}$)	120	350	---
$h_{FE(2)}$	DC Current Gain ($I_C=500\text{mAdc}$, $V_{CE}=1.0\text{Vdc}$)	50	---	---
$V_{CE(sat)}$	Collector-Emitter Saturation Voltage ($I_C=500\text{mAdc}$, $I_B=50\text{mAdc}$)	---	0.6	Vdc
$V_{BE(sat)}$	Base-Emitter Saturation Voltage ($I_C=500\text{mAdc}$, $I_B=50\text{mAdc}$)	---	1.2	Vdc
V_{EB}	Base- Emitter Voltage ($I_E=100\text{mAdc}$)	---	1.4	Vdc

SMALL-SIGNAL CHARACTERISTICS

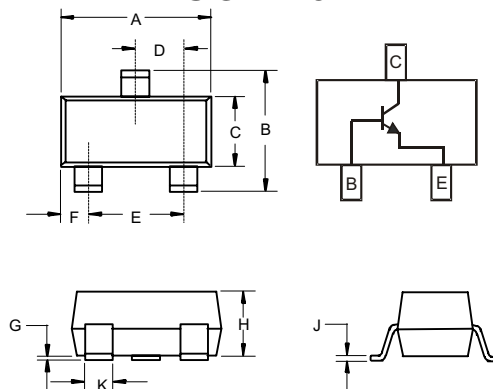
f_T	Transistor Frequency ($I_C=20\text{mAdc}$, $V_{CE}=6.0\text{Vdc}$, $f=30\text{MHz}$)	150	---	MHz
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CLASSIFICATION OF $h_{FE(1)}$

Rank	L	H
Range	120-200	200-350

NPN Silicon Plastic-Encapsulate Transistor

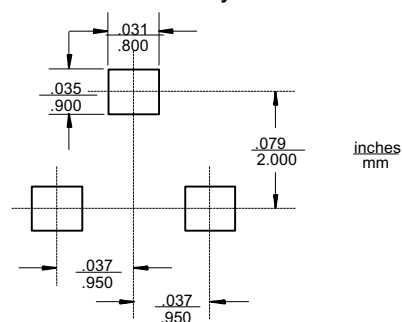
SOT-23



DIMENSIONS

DIM	INCHES		MM		NOTE
	MIN	MAX	MIN	MAX	
A	.110	.120	2.80	3.04	
B	.083	.098	2.10	2.64	
C	.047	.055	1.20	1.40	
D	.035	.041	.89	1.03	
E	.070	.081	1.78	2.05	
F	.018	.024	.45	.60	
G	.0005	.0039	.013	.100	
H	.035	.044	.89	1.12	
J	.003	.007	.085	.180	
K	.015	.020	.37	.51	

Suggested Solder Pad Layout



Ordering Information :

Device	Packing
Part Number-TP	Tape&Reel: 3Kpcs/Reel

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